S.F. for "Output Feedback Stochastic Model Predictive Control for Discrete-time Stochastic Linear System via Block Coordinate Descent"

Appendix A
Matrix Definition

$$\mathcal{A}=\begin{bmatrix} I \\ A \\ A^2 \\ \vdots \\ A^N \end{bmatrix} \in \mathbb{R}^{(N+1)n_x \times n_x},$$

$$\mathcal{K}=\begin{bmatrix} K_0 & 0 & \cdots & \cdots & 0 \\ 0 & K_1 & \ddots & \ddots & \vdots \\ 0 & 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & K_{N-1} & 0 \\ 0 & \cdots & 0 & 0 & 0 \end{bmatrix} \in \mathbb{R}^{(N+1)n_u \times (N+1)n_x}$$

$$\mathcal{B}=\begin{bmatrix} 0 & 0 & \cdots & 0 & 0 \\ B & 0 & \cdots & 0 & 0 \\ AB & B & \ddots & \vdots & 0 \\ \vdots & \ddots & \ddots & 0 & 0 \\ A^{N-1}B & A^{N-2}B & \cdots & B & 0 \end{bmatrix} \in \mathbb{R}^{(N+1)n_x \times (N+1)n_u},$$

$$\mathcal{D}=\begin{bmatrix} 0 & 0 & \cdots & 0 & 0 \\ D & 0 & \cdots & 0 & 0 \\ AD & D & \ddots & \vdots & 0 \\ \vdots & \vdots & \ddots & 0 & 0 \\ A^{N-1}D & A^{N-2}D & \cdots & D & 0 \end{bmatrix} \in \mathbb{R}^{(N+1)n_x \times (N+1)n_w},$$

$$\mathcal{C}=\begin{bmatrix} C & 0 & \cdots & 0 \\ 0 & C & \ddots & \vdots \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \cdots & C \end{bmatrix} \in \mathbb{R}^{(N+1)n_y \times (N+1)n_x}.$$

$$\mathcal{L}=\begin{bmatrix} 0 & 0 & \cdots & 0 & 0 \\ L_0 & 0 & \cdots & 0 & 0 \\ AL_0 & L_1 & \ddots & \vdots & 0 \\ \vdots & \ddots & \ddots & 0 & 0 \\ A^{N-1}L_0 & A^{N-2}L_1 & \cdots & L_{N-1} & 0 \end{bmatrix} \in \mathbb{R}^{(N+1)n_x \times (N+1)n_y}.$$

Furthermore, we introduce the following augmented state vectors:

$$\begin{aligned} X &= \begin{bmatrix} x_0^T & x_1^T & x_2^T & \cdots & x_N^T \end{bmatrix}^T \in \mathbb{R}^{(N+1)n_x}, \\ \widehat{X} &= \begin{bmatrix} \widehat{x}_0^T & \widehat{x}_1^T & \widehat{x}_2^T & \cdots & \widehat{x}_N^T \end{bmatrix}^T \in \mathbb{R}^{(N+1)n_x}, \\ \overline{X} &= \begin{bmatrix} \overline{x}_0^T & \overline{x}_1^T & \overline{x}_2^T & \cdots & \overline{x}_N^T \end{bmatrix}^T \in \mathbb{R}^{(N+1)n_x}, \\ U &= \begin{bmatrix} u_0^T & u_1^T & \cdots & u_{N-1}^T & 1_{n_u}^T \end{bmatrix}^T \in \mathbb{R}^{(N+1)n_u}, \\ W_{(\bullet)} &= \begin{bmatrix} w_{(\bullet)0}^T & w_{(\bullet)1}^T & \cdots & w_{(\bullet)N-1}^T & 1_{n_w}^T \end{bmatrix}^T \in \mathbb{R}^{(N+1)n_w}, \\ V &= \begin{bmatrix} v_0^T & v_1^T & \cdots & v_{N-1}^T & 1_{n_u}^T \end{bmatrix}^T \in \mathbb{R}^{(N+1)n_u}, \end{aligned}$$

$$\begin{aligned} E &= \begin{bmatrix} e_0^T & e_1^T & \cdots & e_N^T \end{bmatrix}^T \in \mathbb{R}^{(N+1)n_x}, \\ F &= \begin{bmatrix} f_0^T & f_1^T & \cdots & f_N^T \end{bmatrix}^T \in \mathbb{R}^{(N+1)n_x}. \end{aligned}$$

$$\mathcal{V}=\left\{ V \;\middle|\; \begin{array}{l} \overline{X}=\mathcal{A}\overline{x}_0+\mathcal{B}V \\ \Sigma_E=\begin{bmatrix}\mathbf{I}-\widetilde{\mathcal{L}}_y^k\mathcal{C} & \widetilde{\mathcal{L}}_y^k\end{bmatrix}\begin{bmatrix} z_e & 0 \\ 0 & \Sigma_{W_2}\end{bmatrix}\begin{bmatrix}\left(\mathbf{I}-\widetilde{\mathcal{L}}_y^k\mathcal{C}\right)^T \\ \left(\widetilde{\mathcal{L}}_y^k\right)^T\end{bmatrix} \\ \Sigma_F=\left(\mathbf{I}+\mathcal{B}\widetilde{\mathcal{K}}^k\right)\begin{bmatrix}\mathbf{I} & \mathbf{I}\end{bmatrix}\begin{bmatrix} z_f & 0 \\ 0 & \Sigma_E\end{bmatrix}\begin{bmatrix}\mathbf{I} \\ \mathbf{I}\end{bmatrix}\left(\mathbf{I}+\mathcal{B}\widetilde{\mathcal{K}}^k\right)^T \\ a_{x,j}^T M_{t,x}\overline{X}+\Psi_{1-\Delta_{x,j}}^{-1} \\ \quad \times\sqrt{a_{x,j}^T M_{t,x} 2\left(\Sigma_E+\Sigma_F\right)\left(a_{x,j}^T M_{t,x}\right)^T} \le b_{x,j} \\ a_{u,l}^T M_{t,u}V+\Psi_{1-\Delta_{u,l}}^{-1} \\ \quad \times\sqrt{a_{u,l}^T M_{t,u}\Sigma_{F,K}\left(a_{u,l}^T M_{t,u}\right)^T} \le b_{u,l} \end{array} \right\}$$

$$\mathcal{Y}=\left\{ \widetilde{\mathcal{L}}_y \;\middle|\; \begin{array}{l} \overline{X}=\mathcal{A}\overline{x}_0+\mathcal{B}V^{k+1} \\ \Sigma_E=\begin{bmatrix}\mathbf{I}-\widetilde{\mathcal{L}}_y\mathcal{C} & \widetilde{\mathcal{L}}_y\end{bmatrix}\begin{bmatrix} z_e & 0 \\ 0 & \Sigma_{W_2}\end{bmatrix}\begin{bmatrix}\mathbf{I}-\widetilde{\mathcal{L}}_y\mathcal{C} & \widetilde{\mathcal{L}}_y\end{bmatrix}^T \\ \Sigma_F=\left(\mathbf{I}+\mathcal{B}\widetilde{\mathcal{K}}^k\right)\begin{bmatrix}\mathbf{I} & \mathbf{I}\end{bmatrix}\begin{bmatrix} z_f & 0 \\ 0 & \Sigma_E\end{bmatrix}\begin{bmatrix}\mathbf{I} \\ \mathbf{I}\end{bmatrix}\left(\mathbf{I}+\mathcal{B}\widetilde{\mathcal{K}}^k\right)^T \\ a_{x,j}^T M_{t,x}\overline{X}+\Psi_{1-\Delta_{x,j}}^{-1} \\ \quad \times\sqrt{a_{x,j}^T M_{t,x} 2\left(\Sigma_E+\Sigma_F\right)\left(a_{x,j}^T M_{t,x}\right)^T} \le b_{x,j} \\ a_{u,l}^T M_{t,u}V^{k+1}+\Psi_{1-\Delta_{u,l}}^{-1} \\ \quad \times\sqrt{a_{u,l}^T M_{t,u}\Sigma_{F,K}\left(a_{u,l}^T M_{t,u}\right)^T} \le b_{u,l} \\ M_{N,x} 2\left(\Sigma_E+\Sigma_F\right) M_{N,x}^T \preccurlyeq \Sigma_{x,N} \end{array} \right\}$$

$$\mathcal{Z}=\left\{ \widetilde{\mathcal{K}} \;\middle|\; \begin{array}{l} \overline{X}=\mathcal{A}\overline{x}_0+\mathcal{B}V^{k+1} \\ \Sigma_E=\begin{bmatrix}\mathbf{I}-\widetilde{\mathcal{L}}_y^{k+1}\mathcal{C} & \widetilde{\mathcal{L}}_y^{k+1}\end{bmatrix}\begin{bmatrix} z_e & 0 \\ 0 & \Sigma_{W_2}\end{bmatrix}\begin{bmatrix}\left(\mathbf{I}-\widetilde{\mathcal{L}}_y^{k+1}\mathcal{C}\right)^T \\ \left(\widetilde{\mathcal{L}}_y^{k+1}\right)^T\end{bmatrix} \\ \Sigma_F=\left(\mathbf{I}+\mathcal{B}\widetilde{\mathcal{K}}\right)\begin{bmatrix}\mathbf{I} & \mathbf{I}\end{bmatrix}\begin{bmatrix} z_f & 0 \\ 0 & \Sigma_E\end{bmatrix}\begin{bmatrix}\mathbf{I} \\ \mathbf{I}\end{bmatrix}\left(\mathbf{I}+\mathcal{B}\widetilde{\mathcal{K}}\right)^T \\ a_{x,j}^T M_{t,x}\overline{X}+\Psi_{1-\Delta_{x,j}}^{-1} \\ \quad \times\sqrt{a_{x,j}^T M_{t,x} 2\left(\Sigma_E+\Sigma_F\right)\left(a_{x,j}^T M_{t,x}\right)^T} \le b_{x,j} \\ a_{u,l}^T M_{t,u}V^{k+1}+\Psi_{1-\Delta_{u,l}}^{-1} \\ \quad \times\sqrt{a_{u,l}^T M_{t,u}\Sigma_{F,K}\left(a_{u,l}^T M_{t,u}\right)^T} \le b_{u,l} \\ M_{N,x} 2\left(\Sigma_E+\Sigma_F\right) M_{N,x}^T \preccurlyeq \Sigma_{x,N} \end{array} \right\}$$

for $t=0,\ldots,N-1$, $j=1,\ldots,m_x$ and $l=1,\ldots,m_u$.

Appendix B
Proof of Theorem 1

Before we proof the Theorem 1, we first provide the following results.

Lemma 2: Let $X \sim \mathcal{N}(\mu_X, \Sigma_X)$, $E \sim \mathcal{N}(0, \Sigma_E)$ and $F \sim \mathcal{N}(0, \Sigma_F)$ be random variables on $\mathbb{R}$. If they satisfy (37) and (11), then $\mu_X = \overline{X}$ and $\Sigma_X \preccurlyeq 2(\Sigma_E + \Sigma_F)$.

Proof: Since $\mu_X = \overline{X}$ is obvious, its proof is omitted.

Define $\Phi = \begin{bmatrix} E^T & F^T \end{bmatrix}^T \in \mathbb{R}^{2(N+1)n_x}$. Since $X = E + F + \overline{X}$, we have

$$\Sigma_X = [\mathbf{I}, \mathbf{I}]\, \Sigma_\Phi\, [\mathbf{I}, \mathbf{I}]^T$$

Furthermore, we have

$$\Sigma_\Phi = \begin{bmatrix} \Sigma_E & \mathrm{cov}(E,F) \\ \mathrm{cov}(E,F)^T & \Sigma_F \end{bmatrix}$$

Since for all $M \succcurlyeq 0$ and all $\begin{bmatrix} z_1^T & z_2^T \end{bmatrix} \neq 0$, we have

$$\begin{aligned} &\begin{bmatrix} -z_1^T & z_2^T \end{bmatrix} \begin{bmatrix} M_{11} & M_{12} \\ M_{12}^T & M_{22} \end{bmatrix} \begin{bmatrix} -z_1 \\ z_2 \end{bmatrix} \\ &= z_1^T M_{11} z_1 + z_2^T M_{22} z_2 \\ &\quad - z_1^T M_{12} z_2 - z_2^T M_{12}^T z_1 \succcurlyeq 0, \end{aligned}$$

Thus, we can obtain that

$$\begin{aligned} &\begin{bmatrix} z_1^T & z_2^T \end{bmatrix} \begin{bmatrix} M_{11} & M_{12} \\ M_{12}^T & M_{22} \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \\ &= z_1^T M_{11} z_1 + z_2^T M_{22} z_2 + z_1^T M_{12} z_2 + z_2^T M_{12}^T z_1 \\ &\leq 2 z_1^T M_{11} z_1 + 2 z_2^T M_{22} z_2 \\ &= \begin{bmatrix} z_1^T & z_2^T \end{bmatrix} \begin{bmatrix} 2M_{11} & 0 \\ 0 & 2M_{22} \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \end{aligned}$$

Since $\Sigma_\Phi \succcurlyeq 0$, we have

$$\begin{bmatrix} \Sigma_E & \mathrm{cov}(E,F) \\ \mathrm{cov}(E,F)^T & \Sigma_F \end{bmatrix} \preceq \begin{bmatrix} 2\Sigma_E & 0 \\ 0 & 2\Sigma_F \end{bmatrix}$$

This implies $\Sigma_X = [\mathbf{I}, \mathbf{I}]\, \Sigma_\Phi\, [\mathbf{I}, \mathbf{I}]^T \preccurlyeq 2\Sigma_E + 2\Sigma_F$. This completes the proof. ■

Based on Lemma 2, we have

$$\begin{aligned} M_{0,x}\mu_X &= M_{0,x}\overline{X} \\ M_{0,x}\Sigma_X M_{0,x}^T &\preceq M_{0,x}\left(2\left(\Sigma_E + \Sigma_F\right)\right) M_{0,x}^T \end{aligned}$$

and

$$\begin{aligned} M_{N,x}\mu_X &= M_{N,x}\overline{X} \\ M_{N,x}\Sigma_X M_{N,x}^T &\preceq M_{N,x}\left(2\left(\Sigma_E + \Sigma_F\right)\right) M_{N,x}^T \end{aligned}$$

Therefore, we can derive the results directly.

Appendix C
Proof of Lemma 1

Proof. The objective function $\widetilde{J}(V, \widetilde{\mathcal{L}}_y, \widetilde{\mathcal{K}})$ is continuously differentiable and bounded below by zero (since $\mathcal{Q} \succcurlyeq 0$ and $\mathcal{R} \succcurlyeq 0$). In Algorithm 1, the sub-problems corresponding to V, $\widetilde{\mathcal{L}}_y$, and $\widetilde{\mathcal{K}}$ are rigorously converted into a strongly convex Quadratic Program (QP), a strongly convex Second-Order Cone Program (SOCP), and another strongly convex SOCP, respectively. Thus, each sub-problem is solved to global optimality for its respective block while fixing the other variables. With the proximal regularization terms $(\sigma/2)\|\cdot\|_F^2$ ($\sigma > 0$), we obtain the sufficient decrease chain:

$$\begin{aligned} &\widetilde{J}(V^{k+1}, \widetilde{\mathcal{L}}_y^{k+1}, \widetilde{\mathcal{K}}^{k+1}) + \frac{\sigma}{2}\left\|\Theta^{k+1} - \Theta^k\right\|_F^2 \\ &\leq \widetilde{J}(V^{k+1}, \widetilde{\mathcal{L}}_y^{k+1}, \widetilde{\mathcal{K}}^k) \leq \widetilde{J}(V^{k+1}, \widetilde{\mathcal{L}}_y^k, \widetilde{\mathcal{K}}^k) \leq \widetilde{J}(V^k, \widetilde{\mathcal{L}}_y^k, \widetilde{\mathcal{K}}^k) \end{aligned}$$

Since the sequence of objective values is non-increasing and bounded below, it converges to a finite infimum and $\lim_{k\to\infty} \|\Theta^{k+1} - \Theta^k\|_F = 0$. Under the strong convexity of the individual QP and SOCP sub-problems, the minimizer for each block is uniquely attained. According to the convergence theory for proximal block coordinate descent methods on semi-algebraic second-order cone programs satisfying the Kurdyka–Łojasiewicz property [33, 57], the entire sequence $\{\Theta^k\}$ has finite length and converges globally to a stationary point. This completes the proof. □

Appendix D
Proof of Theorem 2

Proof. Assume that at time $t-1$, Problem 3 is feasible and yields the converged stationary sequence V_{t-1}^*, $\widetilde{\mathcal{L}}_{y,t-1}^*$, and $\widetilde{\mathcal{K}}_{t-1}^*$ (as guaranteed by Lemma 1). At time t, the controller selects the initial condition according to (80).

Case 1: Problem 3 is feasible with the current estimation $(\hat{\mu}_{x,t}, \hat{\Sigma}_{x,t})$. The initialization strategy directly accepts it, i.e., $(\mu_{x,0|t}, \Sigma_{x,0|t}) = (\hat{\mu}_{x,t}, \hat{\Sigma}_{x,t})$. A feasible solution explicitly exists, rendering the problem trivially feasible.

Case 2: Problem 3 is infeasible with the current estimation. The strategy falls back to the deterministic surrogate bounds optimized from the previous step: $(\mu_{x,0|t}, \Sigma_{x,0|t}) = \left(\overline{x}_{1|t-1}^*, 2\left(\Sigma_{E,1|t-1}^* + \Sigma_{F,1|t-1}^*\right)\right)$. We construct a candidate shifted sequence by appending the terminal gains K_f, L_f, and a nominal terminal input $v_f = 0$:

$$\begin{aligned} V_t^{cand} &= [v_{1|t-1}^*, \ldots, v_{N-1|t-1}^*, 0] \\ K_t^{cand} &= [K_{1|t-1}^*, \ldots, K_{N-1|t-1}^*, K_f] \\ L_t^{cand} &= [L_{1|t-1}^*, \ldots, L_{N-1|t-1}^*, L_f] \end{aligned}$$

By initializing strictly with the surrogate upper bound optimized at $t-1$, the nominal states and surrogate covariance matrices for the first $N-1$ steps perfectly match the optimal tail of the previous iteration. Specifically, for all $i \in \{0, \ldots, N-1\}$, we have the exact index-shifted propagation:

$$\overline{x}_{i|t}^{cand} = \overline{x}_{i+1|t-1}^*, \Sigma_{E,i|t}^{cand} = \Sigma_{E,i+1|t-1}^*, \Sigma_{F,i|t}^{cand} = \Sigma_{F,i+1|t-1}^*$$

Substituting these shifted variables into the tightened surrogate state chance constraints at time t yields:

$$\begin{aligned} &a_{x,j}^T \overline{x}_{i|t}^{cand} + \Psi_{1-\Delta_x}^{-1} \sqrt{2 a_{x,j}^T \left(\Sigma_{E,i|t}^{cand} + \Sigma_{F,i|t}^{cand}\right) a_{x,j}} \\ &= a_{x,j}^T \overline{x}_{i+1|t-1}^* + \\ &\Psi_{1-\Delta_x}^{-1} \sqrt{2 a_{x,j}^T \left(\Sigma_{E,i+1|t-1}^* + \Sigma_{F,i+1|t-1}^*\right) a_{x,j}} \\ &\leq b_{x,j} \end{aligned}$$

which are strictly satisfied since the optimal sequence at $t-1$ was strictly feasible. By the property established in Lemma 2 ($\Sigma_X \preccurlyeq 2\left(\Sigma_E + \Sigma_F\right)$), satisfying these surrogate constraints rigorously guarantees the satisfaction of the actual physical chance constraints:

$$a_{x,j}^T \mu_{X,i|t} + \Psi_{1-\Delta_x}^{-1} \sqrt{a_{x,j}^T \Sigma_{X,i|t} a_{x,j}} \leq b_{x,j}$$

Similar index-shifted inequality propagation holds for the input chance constraints and the terminal covariance constraint

$M_N(\Sigma_X) M_N^T \preccurlyeq \Sigma_{x,N}$. For the final prediction step N, the shifted terminal state enters the terminal invariant set Ω_f. Under Assumption 1, the local laws ensure all probabilistic terminal constraints and covariance bounds are met. Therefore, the constructed shifted sequences constitute a strictly feasible solution. By mathematical induction, the system maintains recursive feasibility. □

Appendix E
Proof of Theorem 3

Proof. Since $\mathcal{Q} \geq 0$ and $\mathcal{R} \geq 0$, we have that $l_{i|t}^m \geq 0$. Then $J^m(\mu_{X,t}, V_t)$ is positive. From the recursive feasibility established previously, $v_{i|t} \in \mathcal{U}$ is bounded and so is $\overline{x}_{i|t} \in \mathcal{X}$. Then $J^m(\mu_{X,t}, V_t)$ is bounded.

When the system utilizes Strategy 2, the initialization strictly follows the prediction, and we have the standard deterministic decrease:

$$\begin{aligned} &J^m(\mu_{X,t}, V_t) - J^m(\mu_{X,t-1}, V_{t-1}) \\ &\leq v_{N|t}^T R v_{N|t} + \overline{x}_{N+1|t-1}^{*T} Q \overline{x}_{N+1|t-1}^* \\ &\quad - v_{0|t-1}^{*T} R v_{0|t-1}^* - \overline{x}_{0|t}^{*T} Q \overline{x}_{0|t}^* \leq 0 \end{aligned}$$

When Strategy 1 is triggered, the initial nominal state incorporates the latest measurement estimate $\hat{\mu}_{x,t}$. The deviation between the measurement estimate and the prior prediction $\overline{x}_{1|t-1}^*$ is governed by the bounded estimation error dynamics characterized by Σ_E. Because the objective function is locally Lipschitz continuous within the compact feasible sets $\mathcal{X}$ and $\mathcal{U}$, there exists a finite positive constant C_w bounded by the noise covariances Σ_{w_1} and Σ_{w_2} such that the cost variation induced by the measurement update is bounded:

$$J^m(\hat{\mu}_{X,t}, V_t) \leq J^m(\overline{x}_{1|t-1}^*, V_{t-1}) + C_w$$

Therefore, the expected value of the Lyapunov function strictly decreases outside a bounded set defined by C_w, establishing that the nominal closed-loop system is practically stable and converges to a bounded region around the origin rather than exactly to zero. □

Appendix F
Proof of Theorem 4

Proof. Let the optimal cost at time t be $J_t^* = J(\Sigma_{X,t}^*, \mu_{X,t}^*, K_t^*, V_t^*)$. At time $t+1$, under recursive feasibility, the shifted sequence $(V_{t+1}^{cand}, K_{t+1}^{cand}, L_{t+1}^{cand})$ with $v_f = 0$ and terminal gains K_f, L_f constitutes a valid feasible solution. Let the cost associated with this candidate sequence be J_{t+1}^{cand}. By optimality, we have:

$$J_{t+1}^* \leq J_{t+1}^{cand} \tag{1}$$

The candidate cost J_{t+1}^{cand} at time $t+1$ is constructed by removing the first stage cost of time t and adding the terminal stage cost. Before addressing the closed-loop measurement update, we first explicitly derive the algebraic upper bound for the terminal stage cost $l_{N+1|t}^{cand} = l(\overline{x}_{N+1|t}^*, v_f, \Sigma_{x_{N+1|t}}^*, K_f)$.

From the cost definition, the terminal stage cost consists of nominal and variance components:

$$\begin{aligned} l_{N+1|t}^{cand} &= \overline{x}_{N+1|t}^{*T} Q \overline{x}_{N+1|t}^* + v_f^T R v_f \\ &\quad + \mathrm{Tr}\left(\Sigma_{x_{N+1|t}}^* Q\right) + \mathrm{Tr}\left(K_f \Sigma_{F,N+1|t}^* K_f^T R\right) \end{aligned}$$

Since $v_f \in \mathcal{U}$ and $\overline{x}_{N+1|t}^* \in \mathcal{X}$ within the terminal invariant set, their nominal costs are bounded by finite maximums: $v_f^T R v_f \leq J_v^{max}$ and $\overline{x}_{N+1|t}^{*T} Q \overline{x}_{N+1|t}^* \leq J_x^{max}$.

From the surrogate terminal constraint and Lemma 2, we have $\Sigma_{x_{N+1|t}}^* \preccurlyeq 2(\Sigma_E + \Sigma_F) \preccurlyeq \Sigma_{x,N}$. Thus, $\mathrm{Tr}\left(\Sigma_{x_{N+1|t}}^* Q\right) \leq \mathrm{Tr}(\Sigma_{x,N} Q)$. From the input chance constraint reformulation over the compact polytopic input set $\mathcal{U} = \{u \in \mathbb{R}^{n_u} \mid a_{u,l}^T u \leq b_{u,l}, l = 1, \ldots, m_u\}$ whose normal matrix $A_u = [a_{u,1}, \ldots, a_{u,m_u}]^T \in \mathbb{R}^{m_u \times n_u}$ has full column rank ($\lambda_{\min}(A_u^T A_u) > 0$), we know:

$$\begin{aligned} &\Psi_{1-\Delta_{u,l}}^{-1} \sqrt{a_{u,l}^T K_f \Sigma_{F,N+1|t}^* K_f^T a_{u,l}} \leq b_{u,l}, \quad l = 1, \ldots, m_u \\ &\implies \mathrm{Tr}\left(K_f \Sigma_{F,N+1|t}^* K_f^T R\right) \leq \frac{\lambda_{\max}(R)}{\lambda_{\min}(A_u^T A_u)} \sum_{l=1}^{m_u} \left(\frac{b_{u,l}}{\Psi_{1-\Delta_{u,l}}^{-1}} \right)^2 \end{aligned}$$

Defining the deterministic terminal upper bound ξ as:

$$\begin{aligned} \xi &= J_v^{max} + J_x^{max} + \mathrm{Tr}(\Sigma_{x,N} Q) \\ &\quad + \frac{\lambda_{\max}(R)}{\lambda_{\min}(A_u^T A_u)} \sum_{l=1}^{m_u} \left(\frac{b_{u,l}}{\Psi_{1-\Delta_{u,l}}^{-1}} \right)^2 \end{aligned} \tag{2}$$

we algebraically guarantee that $l_{N+1|t}^{cand} \leq \xi$.

Unlike a nominal deterministic case, the actual realization of the candidate cost at $t+1$ depends on the closed-loop initialization strategy. If Strategy 1 is triggered, the state undergoes a measurement update, introducing an innovation jump defined as $\delta_t = \hat{x}_t - \overline{x}_{1|t-1}^*$.

Since the actual control input applied is $u_{t-1} = v_{0|t-1}^* + K_{0|t-1}^*(\hat{x}_{t-1} - \overline{x}_{0|t-1}^*)$, and Strategy 1 implies $\overline{x}_{0|t-1}^* = \hat{x}_{t-1}$, the feedback term vanishes, yielding $u_{t-1} = v_{0|t-1}^*$. Substituting the measurement model $y_{t-1} = C x_{t-1} + w_{2,t-1}$ into the observer dynamics, the nominal terms perfectly cancel out, simplifying the innovation jump to:

$$\delta_t = L_{t-1}^* (C e_{0|t-1} + w_{2,t-1}) \tag{3}$$

Given that $e_{0|t-1}$ and $w_{2,t-1}$ are zero-mean and mutually independent, it follows that $\mathbb{E}[\delta_t] = 0$. The covariance of this jump, denoted as Σ_δ, is strictly determined by the observer gain and noise statistics:

$$\Sigma_\delta = \mathbb{E}[\delta_t \delta_t^T] = L_{t-1}^* \left(C \Sigma_{E,0|t-1}^* C^T + \Sigma_{w_2} \right) L_{t-1}^{*T} \tag{4}$$

As the candidate control sequence V_{t+1}^{cand} is open-loop shifted, this zero-mean jump δ_t propagates linearly through the prediction horizon as $A^i \delta_t$. Evaluating its expectation within the nominal quadratic cost J^m, the cross-terms vanish ($\mathbb{E}[\delta_t] = 0$), yielding an explicit closed-form bound Γ for the expected additional innovation cost:

$$\Gamma = \sum_{i=0}^{N} \mathrm{Tr}\left((A^i)^T Q A^i \Sigma_\delta \right) \tag{5}$$

This explicit mathematical formulation is of paramount importance: it not only proves that Γ is a finite bound (restricted by noise variances and the prediction horizon), but also rigorously justifies the necessity of the proposed BCD algorithm. By

jointly optimizing the observer gain L_t iteratively, the algorithm structurally minimizes the quadratic term Σ_δ, thereby systematically suppressing the innovation cost Γ.

Taking the conditional expectation given the information at time t, the expected cost difference is upper bounded by the terminal cost addition (ξ), the initial stage cost reduction, and the explicit innovation bound Γ:

$$\begin{aligned}\mathbb{E}_w[J_{t+1}^{cand}] - J_t^* &\le l_{N+1|t}^{cand} - l(x_{0|t}^*, u_{0|t}^*) + \Gamma \\ &\le \xi + \Gamma - l(x_{0|t}^*, u_{0|t}^*) \end{aligned} \tag{6}$$

Defining the unified closed-loop bound $\overline{\xi} = \xi + \Gamma$, and utilizing the optimality condition $J_{t+1}^* \le J_{t+1}^{cand}$, we obtain:

$$\mathbb{E}_w[J_{t+1}^*] - J_t^* \le \overline{\xi} - l(x_{0|t}^*, u_{0|t}^*) \tag{7}$$

Taking the total expectation and summing this inequality from $t = 0$ to T yields a telescoping sum without requiring the open-loop assumption $v_{i+1|t}^* = v_{i|t+1}^*$:

$$\sum_{t=0}^{T} \mathbb{E}_w(l(x_{0|t}^*, u_{0|t}^*)) \le J_0^* - \mathbb{E}_w(J_{T+1}^*) + (T+1)\overline{\xi} \tag{8}$$

Since $J_{T+1}^* \ge 0$, we drop the $-\mathbb{E}_w(J_{T+1}^*)$ term. Dividing both sides by T and taking the limit as $T \to \infty$, we obtain:

$$\lim_{T\to\infty} \frac{1}{T} \mathbb{E}_w \left(\sum_{t=0}^{T} l(x_{0|t}^*, u_{0|t}^*) \right) \le \overline{\xi} \tag{9}$$

This rigorously proves that the expected average stage cost is strictly bounded for the closed-loop stochastic system, explicitly capturing both terminal constraint bounds and measurement feedback innovations. □

APPENDIX G
SUPPLEMENTARY SIMULATION RESULTS AND EXTENDED ANALYSIS

A. *Offline Terminal Ingredients and 300-Run Initialization Sensitivity*

For the numerical example in Section V-A with open-loop unstable plant matrices

$$A = \begin{bmatrix} 1.02 & 0.95 \\ 0.06 & 1.02 \end{bmatrix}, \quad B = \begin{bmatrix} 0.85 \\ 1.00 \end{bmatrix}, \quad D = I_2, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix},$$

solving the discrete algebraic Riccati equations and discrete Lyapunov equation offline for (A, B, Q, R) and $(A^T, C^T, D\Sigma_{w_1} D^T, \Sigma_{w_2})$ with $Q = I_2$, $R = 0.1$, $\Sigma_{w_1} = \mathrm{diag}(0.008, 0.012)$, and $\Sigma_{w_2} = 0.008$ yields the stabilizing terminal gains, steady-state covariance bound, and invariant ellipsoid required by Assumption 1, Remark 10, and terminal constraints (53)–(55):

$$\begin{aligned} K_f &= \begin{bmatrix} -0.6365 & -1.0317 \end{bmatrix}, \quad L_f = \begin{bmatrix} 1.3477 \\ 0.5752 \end{bmatrix}, \\ \rho(A + BK_f) &= 0.3679 < 1, \quad \rho(A - L_f C) = 0.3940 < 1, \\ \Sigma_{x,\mathrm{bound}} &= 2\left(\Sigma_{e,\infty} + \Sigma_{f,\infty}\right) = \begin{bmatrix} 0.3260 & 0.0636 \\ 0.0636 & 0.1842 \end{bmatrix}, \\ \Sigma_{x,N} &= 0.85\, M_{N,x} 2\left(\Sigma_{E,\mathrm{ss}} + \Sigma_{F,\mathrm{ss}}\right) M_{N,x}^T, \\ P_f &= \begin{bmatrix} 1.7398 & 0.1280 \\ 0.1280 & 1.1157 \end{bmatrix}, \\ \Omega_f &= \left\{ \overline{x} \in \mathbb{R}^2 : \overline{x}^T P_f \overline{x} \le 2.798 \right\}, \end{aligned}$$

with tightened terminal bounds $b_{x,\mathrm{tight}} = 1.6388$ and $b_{u,\mathrm{tight}} = 1.7620$. Table I summarizes the sensitivity of Algorithm 1 to initial covariance scaling $(\Sigma_E^0, \Sigma_F^0) = cI$ across $c \in \{0.1, 0.5, 1.5, 3.0\}$ and across 300 initializations spanning $c \in [0.05, 2.0]$. Across all scaling regimes, the BCD algorithm converges in 5 outer iterations to $J_0^* = 19.3885$ in Case 1 and to $J_0^* = 23.7853 \pm 0.0039$ with a relative standard deviation of 0.016% in Case 2.

TABLE I
SENSITIVITY OF BCD CONVERGENCE TO INITIAL COVARIANCE SCALING $(\Sigma_E^0, \Sigma_F^0) = cI$ $(N = 8)$

Initialization Scale $(\Sigma_E^0, \Sigma_F^0) = cI$	Case 1 / 2 BCD Iters	Case 1 Initial J_0^0	Case 1 Final J_0^*	Case 2 Initial J_0^0	Case 2 Final J_0^*
$c = 0.10$	5 / 5	185.82	19.3885	100.30	23.7848
$c = 0.50$ (Default)	5 / 5	211.50	19.3885	125.98	23.7874
$c = 1.50$	5 / 5	275.70	19.3885	190.18	23.7843
$c = 3.00$	5 / 5	372.00	19.3885	286.48	23.7799
300 Runs ($c \in [0.05, 2.0]$)	5 / 5	—	19.3885	—	$23.7853 \pm 3.9 \times 10^{-3}$

B. *Quantitative Gain-Strategy Ablation & Step-by-Step Recursion Verification*

Table II provides the quantitative comparison of the four gain-design strategies depicted in Figs. 1 and 2 of the main paper over 100 Common-Random-Number Monte Carlo realizations. Here, the constant gains $(L_{\mathrm{fix}}, K_{\mathrm{fix}}) = (0.30L_{\mathrm{ss}}, 0.22K_{\mathrm{lqr}})$ are nominal stabilizing gains with Schur spectral radii $\rho(A + BK_{\mathrm{fix}}) = 0.8804$ and $\rho(A - L_{\mathrm{fix}}C) = 0.8573$, which ensures that $2\Sigma_{E,N}(L_{\mathrm{fix}}) \prec \Sigma_{x,N}$ so all four ablation strategies remain strictly feasible under the offline terminal covariance bounds (53)–(55). Furthermore, because the $(N+1)$-block prediction matrices retain the initial time step $t = 0$ with dimensions $p_{\mathrm{full}} = (N+1)n_x$, $p_{y,\mathrm{full}} = (N+1)n_y$, and $\mu = Nn_u$, they map bijectively to the causal step-by-step innovation and disturbance-feedback gains. Consequently, the block-matrix propagation in (8a)–(8f) and the step-by-step causal state-space recursion yield identical trajectories and terminal covariances up to machine precision below 10^{-14}.

TABLE II
QUANTITATIVE ABLATION OF GAIN-DESIGN STRATEGIES OVER 100 COMMON-RANDOM-NUMBER TRIALS IN CASE 1 AND CASE 2 $(N = 8)$

Case / Gain Strategy	Surrogate Cost J_0	Empirical Cost $J_{\mathrm{emp}} \pm \mathrm{CI}_{95}$	Terminal Cov. $\mathrm{Tr}(\Sigma_{X,N}^{\mathrm{emp}} / \Sigma_{X,N}^{\mathrm{true}})$	Tracking RMSE	State / Input Viol. (%)	Solve Time (ms)
Case 1 ($\Delta_u = 0.10$)						
Proposed (TV L_t, TV K_t)	**19.39**	**18.06 ± 0.38**	**0.308 / 0.287**	**0.421**	**0.0% / 0.0%**	19.09
TV K_t, Fixed L_t	20.84	18.76 ± 0.56	0.473 / 0.427	0.488	0.0% / 0.0%	21.60
TV L_t, Fixed K_t	24.66	18.86 ± 0.66	0.427 / 0.397	0.492	0.0% / 0.0%	8.29
Fixed L_t and Fixed K_t	23.53	21.61 ± 1.53	1.713 / 1.618	0.679	0.0% / 0.0%	0.25
Case 2 ($\Delta_x = \Delta_u = 0.20$)						
Proposed (TV L_t, TV K_t)	**23.79**	**22.38 ± 0.41**	**0.284 / 0.287**	**0.400**	**4.0% / 0.0%**	37.94
TV K_t, Fixed L_t	25.25	22.89 ± 0.53	0.352 / 0.427	0.446	4.0% / 0.0%	26.61
TV L_t, Fixed K_t	29.33	23.11 ± 0.56	0.345 / 0.391	0.446	3.0% / 0.0%	17.30
Fixed L_t and Fixed K_t	28.15	25.24 ± 1.21	1.235 / 1.618	0.604	3.0% / 0.0%	3.90

C. *Robustness Against Non-Gaussian Disturbances*

Table III evaluates the proposed BCD-SMPC in Case 2 ($N = 8$, 200 Monte Carlo trials) across four zero-mean, covariance-matched disturbance distributions: (i) Gaussian $\mathcal{N}(0, \Sigma_w)$ with $\Psi_{1-\Delta}^{-1} = 0.8416$, (ii) Uniform $\mathcal{U}[-\sqrt{3}\sigma, \sqrt{3}\sigma]$, (iii) heavy-tailed Laplace $\mathrm{Laplace}(0, \sigma/\sqrt{2})$, and (iv) heavy-tailed Student-t ($\nu = 5$) with Cantelli bound $\kappa_{\mathrm{DR}} = 2.0$. In all

cases, the empirical violation rates remain strictly below the 20% risk threshold.

TABLE III
CLOSED-LOOP ROBUSTNESS UNDER NON-GAUSSIAN DISTURBANCES IN CASE 2 ($N = 8$, $\Delta_x = \Delta_u = 20\%$)

Disturbance Distribution	Tightening Factor κ_x	Empirical Cost J_{emp}	Terminal $\text{Tr}(\Sigma^{\text{emp}}_{X,N})$	State Viol. Rate (%)	Input Viol. Rate (%)
Gaussian $\mathcal{N}(0, \Sigma_w)$	Gaussian (0.8416)	22.29	0.285	2.5%	0.0%
Uniform $\mathcal{U}[-\sqrt{3}\sigma, \sqrt{3}\sigma]$	Cantelli (2.0000)	26.32	0.324	0.0%	0.0%
Heavy-Tailed Laplace	Cantelli (2.0000)	26.40	0.352	0.5%	0.0%
Heavy-Tailed Student-t ($\nu = 5$)	Cantelli (2.0000)	26.21	0.259	0.0%	0.0%

D. *Control Experiment With Unscaled Steady-State DARE Gains ($L_{base} = L_{ss}$, $K_{base} = K_{lqr}$)*

To verify that our conclusions do not rely on the nominal constant-gain scaling $(L_{\text{base}}, K_{\text{base}})$, Table IV reports a control experiment in which all six competing baselines use the unscaled infinite-horizon DARE steady-state Kalman gain $L_{\text{base}} = L_{\text{ss}}$ and LQR gain $K_{\text{base}} = K_{\text{lqr}}$. Even compared against unscaled infinite-horizon DARE gains, the proposed BCD-SMPC achieves competitive constraint-compliant empirical cost of 22.04 in Case 2 and the lowest cost of 27.98 on the 6-DOF vehicle, while reducing the terminal covariance trace on the 6-DOF vehicle by 2.27–2.72 times from 0.052–0.063 down to 0.023 with 0.0% constraint violations.

TABLE IV
BASELINE CONTROL EXPERIMENT WITH UNSCALED INFINITE-HORIZON DARE GAINS ($L_{\text{BASE}} = L_{\text{SS}}$, $K_{\text{BASE}} = K_{\text{LQR}}$)

Benchmark / Method	Empirical Cost $J_{\text{emp}} \pm \text{CI}_{95}$	$\text{Tr}(\Sigma^{\text{emp}}_{X,N}/\Sigma^{\text{true}}_{X,N})$	Tracking RMSE	State / Input Viol. (%)
Case 2 ($N = 6$)				
Proposed BCD-SMPC	**22.04 ± 0.43**	0.301 / 0.313	0.409	**3.0% / 0.0%**
CS-SMPC (L_{ss})	21.03 ± 0.35	0.232 / 0.251	0.401	19.0% / 0.0%
Tube-MPC (L_{ss}, K_{lqr})	21.44 ± 0.37	0.232 / 0.253	0.400	6.0% / 0.0%
RF-MPC (L_{ss}, K_{lqr})	22.13 ± 0.39	0.232 / 0.253	0.400	3.0% / 0.0%
DR-MPC (L_{ss})	22.31 ± 0.43	0.306 / 0.322	0.405	3.0% / 0.0%
DDRMPC (L_{ss}, K_{lqr})	22.43 ± 0.41	0.232 / 0.253	0.400	2.0% / 0.0%
SDP-MPC (L_{ss})	22.49 ± 0.40	0.231 / 0.253	0.403	2.0% / 0.0%
Case 4 (6-DOF Vehicle, $N = 12$)				
Proposed BCD-SMPC	**27.98 ± 5.00**	**0.023 / 0.023**	**0.287 m**	**0.0% / 0.0%**
Tube-MPC (L_{ss}, K_{lqr})	32.04 ± 5.12	0.063 / 0.072	0.297 m	0.0% / 0.0%
CS-SMPC (L_{ss})	32.40 ± 5.14	0.062 / 0.072	0.294 m	0.0% / 3.0%
SDP-MPC (L_{ss})	48.85 ± 5.06	0.052 / 0.057	0.345 m	0.0% / 4.0%
DDRMPC (L_{ss}, K_{lqr})	55.37 ± 5.65	0.063 / 0.072	0.297 m	0.0% / 0.0%
RF-MPC (L_{ss}, K_{lqr})	102.05 ± 6.90	0.063 / 0.072	0.297 m	0.0% / 0.0%
DR-MPC (L_{ss})	264.69 ± 15.02	0.173 / 0.196	0.342 m	0.0% / 0.0%

E. *Complete 6-DOF AutoRally Racing Vehicle Simulation Protocol, Jacobian Matrices, and Fair Baseline Comparison*

To ensure complete transparency and reproducibility of Section V-C without any ambiguity, we detail the exact mathematical model, discretization, track geometry, constraint tightening, and baseline evaluation protocol implemented in `bcd_smpc.py` and `run_all.py`:

1) State Vector, Equilibrium, and Exact ZOH Discretization: The 6-DOF curvilinear dynamic bicycle tracking-error state is ordered as $\tilde{x} = [e_y,\, v_y,\, e_\psi,\, w_z,\, \tilde{v}_x,\, \tilde{s}]^T \in \mathbb{R}^6$ with control input $u = [\delta,\, a]^T \in \mathbb{R}^2$ and physical parameters listed in Table V. The front and rear static normal tire loads are $f_{Fz} = mg\ell_R/(\ell_F + \ell_R) = 87.09$ N and $f_{Rz} = mg\ell_F/(\ell_F + \ell_R) = 128.73$ N, giving peak Pacejka lateral forces $D_f = f_{Fz}\,\text{tire}_D = 95.80$ N and $D_r = f_{Rz}\,\text{tire}_D = 141.61$ N. Along the reference racing line with $v_x = v_g = 9.0$ m/s, $\rho(s) = \rho_0 + \rho' \tilde{s}$, $\rho_0 = 0.25\ \text{m}^{-1}$, $\rho' = 0.05\ \text{m}^{-2}$, and nominal steady-state yaw rate $w_{z,\text{ref}} = \rho_0 v_g$, the origin $\tilde{x}_{\text{eq}} = \mathbf{0}, u_{\text{eq}} = \mathbf{0}$ is an exact equilibrium satisfying $f_{\text{veh}}(\mathbf{0}, \mathbf{0}) = \mathbf{0}$ identically. Evaluating the continuous-time Jacobian matrices $A_c = \left.\frac{\partial f_{\text{veh}}}{\partial \tilde{x}}\right|_{(\mathbf{0},\mathbf{0})}$ and $B_c = \left.\frac{\partial f_{\text{veh}}}{\partial u}\right|_{(\mathbf{0},\mathbf{0})}$ gives:

$$A_c = \begin{bmatrix} 0 & 1 & 9.00 & 0 & 0 & 0 \\ 0 & -4.67 & 0 & -9.00 & 0 & 0 \\ -0.56 & 0 & 0 & 1 & -0.25 & -0.45 \\ 0 & 0 & 0 & -7.30 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 2.25 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}, \quad B_c = \begin{bmatrix} 0 & 0 \\ 16.96 & 0 \\ 0 & 0 \\ 115.33 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

The discrete-time matrices A and B with sampling period $\Delta t = 0.12$ s and horizon $N = 12$ are obtained via exact zero-order-hold matrix exponentiation:

$$\begin{bmatrix} A & B \\ \mathbf{0}_{2\times 6} & \mathbf{I}_2 \end{bmatrix} = \exp\left(\begin{bmatrix} A_c & B_c \\ \mathbf{0}_{2\times 6} & \mathbf{0}_{2\times 2} \end{bmatrix} \Delta t \right).$$

2) Partial-Observation Model, Noise Statistics, and Polytopic Constraints: Only lateral error e_y, heading error e_ψ, and longitudinal speed error $\tilde{v}_x$ are measured with $n_y = 3 < n_x = 6$, while lateral slip velocity v_y, yaw rate w_z, and arc-length drift $\tilde{s}$ are unmeasured:

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}, \quad A_x = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}, \quad b_x = \begin{bmatrix} 0.65 \\ 1.80 \\ 1.20 \end{bmatrix},$$

and input box constraints $A_u = [\mathbf{I}_2,\, -\mathbf{I}_2]^T \in \mathbb{R}^{4\times 2}$, $b_u = [0.50,\, 2.50,\, 0.50,\, 2.50]^T$ with $\Delta_x = \Delta_u = 0.10$. Under uniform Bonferroni risk allocation, $\Delta_{x,j} = \Delta_x/3 = 0.0333$ gives $\Psi^{-1}_{1-\Delta_{x,j}} = 1.8339$ and $\Delta_{u,l} = \Delta_u/4 = 0.0250$ gives $\Psi^{-1}_{1-\Delta_{u,l}} = 1.9600$ across $t = 1, \ldots, N$.

3) S-Bend Chicane Centerline Mapping and Closed-Loop Evaluation: In Figs. 6–7, the S-bend dirt chicane centerline curvature profile $\kappa_{\text{ref}}(s) = 0.23\exp(-((s-6.5)/3.6)^2) - 0.23\exp(-((s-19.5)/3.6)^2)$ is integrated via cumulative trapezoidal quadrature to obtain the reference heading $\psi_{\text{ref}}(s)$ and Cartesian centerline $(X_{\text{ref}}(s), Y_{\text{ref}}(s))$ with unit normal vector $n(s) = [-\sin\psi_{\text{ref}}(s),\, \cos\psi_{\text{ref}}(s)]^T$. Both the nominal trajectory and each simulated sample trajectory $(e^{(s)}_{y,t}, \tilde{s}^{(s)}_t)$, together with the analytical 3σ position uncertainty ellipses with semi-axes $3\sqrt{\Sigma_{x,t}[1,1]}$ and $3\sqrt{\Sigma_{x,t}[6,6]}$, are mapped onto global track coordinates via $[X_t, Y_t]^T = [X_{\text{ref}}(s_{\text{node},t} + 2\tilde{s}_t), Y_{\text{ref}}(s_{\text{node},t} + 2\tilde{s}_t)]^T + 2e_{y,t}\, n(s_{\text{node},t} + 2\tilde{s}_t)$. All methods are evaluated over the same 100 Common-Random-Number trials with fixed random seed 400, both on the discrete-time linearized model and by integrating the continuous-time nonlinear Pacejka ODEs via 4th-order Runge–Kutta without drift subtraction.

4) Fair Baseline Protocol and Physical Interpretation of Figs. 6(a)–(b): All six competing baselines and the fixed-gain ablations share the Schur-stable observer gain $L_{\text{base}} = L_{\text{fix}} = 0.70 L_{\text{ss}}$ with $\rho(A - L_{\text{fix}}C) = 0.9671 < 1$ and nominal feedback gain $K_{\text{base}} = K_{\text{fix}} = 0.65 K_{\text{lqr}}$ with $\rho(A + BK_{\text{fix}}) = 0.8476 < 1$, while Table IV reports the control experiment with unscaled $(L_{\text{ss}}, K_{\text{lqr}})$. Because $n_y = 3 < n_x = 6$ leaves $[v_y, w_z, \tilde{s}]^T$ unmeasured, the steady-state observer mode with $\rho(A - L_{\text{ss}}C) = 0.9528$ contracts much more slowly than state feedback with $\rho(A + BK_{\text{lqr}}) = 0.8684$, and estimation error Σ_E directly excites $\Sigma_F = (\mathbf{I} + \mathcal{B}\widetilde{\mathcal{K}})(z_f + \Sigma_E)(\mathbf{I} + \mathcal{B}\widetilde{\mathcal{K}})^T$. Consequently, optimizing time-varying L_t in both Fig. 6(a) and

Fig. 6(b) reduces $\mathrm{Tr}(\Sigma_{E,N})$ from 0.0221 to 0.0048–0.0057, which explains why Figs. 6(a)–(b) achieve $\mathrm{Tr}(\Sigma^{\mathrm{emp}}_{X,N}) = 0.023$ and 0.026 compared with 0.073 and 0.083 for the Fixed-L_t strategies in Figs. 6(c)–(d). Jointly optimizing time-varying K_t in Fig. 6(a) further reduces the velocity and yaw-rate tracking variances so that $\mathrm{Tr}(\Sigma_{F,N})$ drops from 0.0126 to 0.0098 and surrogate $\mathrm{Tr}(2(\Sigma_{E,N}+\Sigma_{F,N}))$ drops from 0.0367 to 0.0293, satisfying terminal constraint (55) with margin $+0.0548$ whereas TV L_t Fixed K_t violates (55) with margin -0.8468. In the warm-start benchmark, the proposed BCD-SMPC converges in 3 iterations and 61.46 ms under cold start, in 2 iterations and 57.44 ms under offline LQG warm start at $t = 0$, and in 2.0 iterations and 28.44 ms mean step time with zero surrogate constraint violations across receding-horizon steps $t = 1, \ldots, 15$.

TABLE V
PHYSICAL PARAMETERS OF THE 6-DOF AUTORALLY OFF-ROAD RACING VEHICLE MODEL

Parameter	Symbol	Value
Vehicle mass / Yaw moment of inertia	m / I_z	22.0 kg / 1.10 kg $\cdot$ m^2
Front / Rear CG-to-axle distance	ℓ_F / ℓ_R	0.34 m / 0.23 m
Front / Rear static normal tire load	f_{Fz} / f_{Rz}	87.09 N / 128.73 N
Pacejka stiffness / shape / peak factors	$B_{f,r}$ / $C_{f,r}$ / tire$_D$	4.10 / 0.95 / 1.10
Target longitudinal speed / Chicane curvature	v_g / ρ_0 / ρ'	9.0 m/s / 0.25 m^{-1} / 0.05 m^{-2}
Sampling period / Prediction horizon	Δt / N	0.12 s / 12
State / Input / Measured output dimensions	n_x / n_u / n_y	6 / 2 / 3